

Factoring Cubes and Higher Degrees

Answer Key

Algebra 2

Quick Answers

- | | |
|-------------------------------------|---|
| 1. $(x - 3)(x^2 + 3x + 9)$ | $(x - 1)(x + 1)(x - 2)(x + 2)$, (b) the |
| 2. $(2x + 5)(4x^2 - 10x + 25)$ | x-intercepts = 1, -1, 2, -2, — |
| 3. $(x + 4)(x - 3)(x + 3)$ | 8. $(4x - 1)(16x^2 + 4x + 1)$ |
| 4. $(x - 2)(x + 2)(x - 3)(x + 3)$ | 9. $(x - 2)(x^2 + 2x + 4)(x + 2)(x^2 - 2x + 4)$ |
| 5. $2(x - 2)(x^2 + 2x + 4)$ | 10. (a) the complete factorization = |
| 6. $(3x - 2)(x - 2)(x + 2)$ | $(x + 2)^2(x - 2)$, (b) the solutions = -2, 2, — |
| 7. (a) the complete factorization = | |
-

What is verified

12 of 14 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-APR.B.3 – problems 7, 10

HSA-SSE.B.3 – problems 1, 2, 3, 4, 5, 6, 8, 9

Difficulty 1–5 of 5 across 14 tagged checks.

1. Factor completely: $x^3 - 27$.

Solution: Two terms, and both are perfect cubes: x^3 and 3^3 . The difference-of-cubes pattern with $a = x$ and $b = 3$ gives $(a - b)(a^2 + ab + b^2)$. The trailing quadratic $x^2 + 3x + 9$ has discriminant $9 - 36 < 0$, so it does not factor further.

$(x - 3)(x^2 + 3x + 9)$

2. Factor completely: $8x^3 + 125$.

Solution: $8x^3 = (2x)^3$ and $125 = 5^3$, so this is a sum of cubes with $a = 2x$ and $b = 5$: $(a + b)(a^2 - ab + b^2)$. That gives $a^2 = 4x^2$, $ab = 10x$ and $b^2 = 25$.

$(2x + 5)(4x^2 - 10x + 25)$

3. Factor completely: $x^3 + 4x^2 - 9x - 36$.

Solution: Four terms, so group them in pairs:

$$x^2(x + 4) - 9(x + 4) = (x + 4)(x^2 - 9)$$

The second factor is a difference of squares, so it breaks down again — this is where "factor completely" earns its name.

$$(x + 4)(x - 3)(x + 3)$$

4. Factor completely: $x^4 - 13x^2 + 36$.

Solution: Treat x^2 as a single letter: two numbers with product 36 and sum -13 are -4 and -9 , so the expression is $(x^2 - 4)(x^2 - 9)$. Both are differences of squares.

$$(x - 2)(x + 2)(x - 3)(x + 3)$$

5. Factor completely: $2x^3 - 16$.

Solution: Neither $2x^3$ nor 16 is a perfect cube, but the common factor 2 comes out first and leaves one that is: $2(x^3 - 8)$, and $8 = 2^3$. Applying the difference of cubes with $a = x$, $b = 2$:

$$2(x - 2)(x^2 + 2x + 4)$$

6. Factor completely: $3x^3 - 2x^2 - 12x + 8$.

Solution: Group in pairs; the shared bracket is $3x - 2$:

$$x^2(3x - 2) - 4(3x - 2) = (3x - 2)(x^2 - 4)$$

and $x^2 - 4$ is a difference of squares.

$$(3x - 2)(x - 2)(x + 2)$$

7. $x^4 - 5x^2 + 4$. (a) Factor completely. (b) x -intercepts. (c) Sketch.

Solution (a): As a quadratic in x^2 : product 4, sum -5 , so -1 and -4 , giving $(x^2 - 1)(x^2 - 4)$. Both are differences of squares.

$$(x - 1)(x + 1)(x - 2)(x + 2)$$

Solution (b): Each factor set to zero gives one intercept.

$$x = 1, -1, 2, -2$$

Part (c) — instructor-judged. The sketch must cross the x -axis at those four values and nowhere else, and must rise on both far ends (even degree, positive leading coefficient). Between -1 and 1 the curve is above the axis (at $x = 0$, $y = 4$); between 1 and 2 , and between -2 and -1 , it dips below. Accept any smooth freehand curve with those five features; do not require accurate heights. Grade the sketch against the intercepts the student listed in part (b).

8. Factor completely: $64x^3 - 1$.

Solution: $64x^3 = (4x)^3$ and $1 = 1^3$, so this is a difference of cubes with $a = 4x$ and $b = 1$: $a^2 = 16x^2$, $ab = 4x$, $b^2 = 1$.

$$(4x - 1)(16x^2 + 4x + 1)$$

9. Factor completely: $x^6 - 64$.

Solution: $x^6 = (x^3)^2$ and $64 = 8^2$, so start with the difference of squares:

$$x^6 - 64 = (x^3 - 8)(x^3 + 8)$$

Now each factor is a cube pattern: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$ and $x^3 + 8 = (x + 2)(x^2 - 2x + 4)$. Taking the squares route first is what gets all four factors; taking cubes first ($x^6 - 64 = (x^2)^3 - 4^3$) stops one step short.

$(x - 2)(x^2 + 2x + 4)(x + 2)(x^2 - 2x + 4)$

10. $x^3 + 2x^2 - 4x - 8$. (a) Factor. (b) Solve. (c) Which zero does the graph touch, and why?

Solution (a): Group in pairs: $x^2(x + 2) - 4(x + 2) = (x + 2)(x^2 - 4)$, and $x^2 - 4 = (x - 2)(x + 2)$, so the factor $x + 2$ appears twice.

$(x + 2)^2(x - 2)$

Solution (b): Setting each factor to zero gives $x = -2$ (from the repeated factor) and $x = 2$.

$x = -2, 2$

Part (c) — instructor-judged. A full-credit explanation identifies $x = -2$ and says it comes from a factor raised to the power 2, so it has multiplicity two, and an even multiplicity makes the curve touch the axis and turn back rather than pass through. A student who tests the sign of the expression just either side of -2 and observes that it does not change has demonstrated it. Naming the repeated factor with no statement about the graph earns nothing.